

Laboratorio #7

1. $S \rightarrow 0A011B11BB$
 $A \rightarrow C \quad C \rightarrow S1E$
 $B \rightarrow S1A$

1. Eliminar epsilon

$S \rightarrow 0011B11BB10A0$
 $S \rightarrow 0A01111B11B$
 $S \rightarrow 0A010011B1111BB1B$
 $A \rightarrow \epsilon \quad C \rightarrow S$
 $B \rightarrow S1A$

2. Eliminar producciones unitarias

$P_0 = \{(S,S), (A,A), (B,B), (C,C)\}$

$(S,S) \quad S \rightarrow B \quad (S,B)$

$(A,A) \quad A \rightarrow \epsilon \quad (A,C)$

$(S,B) \quad B \rightarrow A \quad (S,A)$

$(A,C) \quad C \rightarrow S \quad (A,S)$

$(S,A) \quad A \rightarrow C \quad (S,C)$

$(S,C) \quad C \rightarrow S \quad (S,S)$

$(C,C) \quad C \rightarrow S \quad (C,S)$

$(B,B) \quad B \rightarrow A \quad (B,A)$

$(S,S) \quad S \rightarrow 0A011B11001111BB$

$(B,A) \quad A \rightarrow C \quad (B,C)$

$(S,A), (S,B), (S,C)$ no generan

$(B,C) \quad C \rightarrow S \quad (B,S)$

$(B,S) \quad S \rightarrow 0A011B11001111BB$

$(B,B), (B,A), (B,C)$ no generan

$S \rightarrow 0A011B11001111BB$

$A \rightarrow \quad "$

$(A,S) \quad S \rightarrow 0A011B11001111BB$

$B \rightarrow \quad "$

$(A,A), (A,B), (A,C)$ no generan

$C \rightarrow \quad "$

$(C,S) \quad S \rightarrow 0A011B11001111BB$

$(C,C), (C,B), (C,A)$ no genera

eliminar:

$T = \{0, 1\}$

$Y_1 = \{S\}$

$S \rightarrow 0A011B11001111BB$

$W_1 = \{S, A, B, C\}$

$Y_2 = \{S, A, B, 0, 1\}$

$A \rightarrow 0A011B11001111BB$

$W_2 = \{S, A, B, C\}$

$Y_3 = \{S, A, B, 0, 1\}$

$B \rightarrow 0A011B11001111BB$

Chomsky:

$C_1 = AT_0 \quad C_2 = BT_1$

$T_0 = 0 \quad T_1 = 1$

$S \rightarrow T_0 C_1 | T_1 C_2 | 001111BB$

$S \rightarrow T_0 AT_0 | T_1 BT_1 | 001111BB$

$A \rightarrow T_0 C_1 | T_1 C_2 | 001111BB$

$A \rightarrow T_0 AT_0 | T_1 BT_1 | 001111BB$

$B \rightarrow T_0 C_1 | T_1 C_2 | 001111BB$

$B \rightarrow T_0 AT_0 | T_1 BT_1 | 001111BB$

2. $S \rightarrow aAa | bBb | E$

$A \rightarrow C | a$

$B \rightarrow C | b$

$C \rightarrow CDE | E$

$D \rightarrow A | B | ab$

1. Eliminar epsilon

$S \rightarrow aAa | aal | bBb |$

$S \rightarrow aAa | bbb | bBb$

$S \rightarrow aAa | aal | bbb | bBb$

$A \rightarrow C | a$

$B \rightarrow C | b$

$C \rightarrow CDE | DE | CE | E$

$D \rightarrow A | B | ab$

2. Eliminar unitarios

$P_0 = \{(S, S), (A, A), (B, B), (C, C), (D, D)\}$

$(C, C) \ C \rightarrow E \ (C, E)$

(S, S)

$(A, A) \ A \rightarrow C \ (A, C)$

$(A, C) \ C \rightarrow E \ (A, E)$

$(B, B) \ B \rightarrow C \ (B, C)$

$(B, C) \ C \rightarrow E \ (B, E)$

$(D, D) \ D \rightarrow A \ (D, A)$

$(D, A) \ A \rightarrow C \ (D, C)$

$(D, C) \ C \rightarrow E \ (D, E)$

$(D, D) \ D \rightarrow B \ (D, B)$

(D, B)

$(C, C): CDE | DE | CE$

$(C, E): \text{no produce}$

$(S, S): aAa | aal | bbb | bBb$

$(A, A): a$

$(A, C): CDE | DE | CE$

$(B, B): b$

$(B, C): CDE | DE | CE$

$(D, D): ab$

$(D, A): a$

$(D, B): b$

$(D, C): CDE | DE | CE$

$(D, E): -$

$S \rightarrow aAa | aal | bbb | bBb$

$A \rightarrow CDE | DE | CE | a$

$B \rightarrow CDE | DE | CE | b$

$C \rightarrow CDE | DE | CE$

$D \rightarrow CDE | DE | CE | ab | a | b$

3. Eliminar innecesarios:

$T: \{a, b\}$

$W_1: \{S, A, B, D, a, b\}$

$W_2: \{S, A, B, C, D, a, b\}$

$W_3: \{S, A, B, C, D, a, b\}$

$Y_1: \{S\}$

$Y_2: \{D, A, B, a, b, S\}$

$Y_3: \{D, A, B, a, b, C, S\}$

$S \rightarrow$

$A \rightarrow$

$B \rightarrow$

$C \rightarrow$

$D \rightarrow$

4. forma chomsky

$C_1 = DE \quad T_1 = a \quad T_2 = b$

$S \rightarrow T_1 A T_1 | T_1 T_1 | T_2 E | T_2 B T_2$

$A \rightarrow C C_1 | DE | CE | T_1$

$B \rightarrow C C_1 | DE | CE | T_2$

$C \rightarrow C C_1 | DE | CE |$

$D \rightarrow C C_1 | DE | CE | T_1 T_2 | T_1 T_2$

$C_2 = A T_1 \quad C_3 = B T_1$

$S \rightarrow T_1 C_1 | T_1 T_1 | T_2 T_2 | T_2 C_3$

$A \rightarrow C C_1 | DE | CE | T_1$

$B \rightarrow C C_1 | DE | CE | T_2$

$C \rightarrow C C_1 | DE | CE |$

$D \rightarrow C C_1 | DE | CE | T_1 T_2 | T_1 T_2$

3. $S \rightarrow ASA|aB$ 1 eliminar epsilon

$A \rightarrow B|S$

$B \rightarrow b|E$

$S \rightarrow ASA|a|aB$

$S \rightarrow SA|AS|ASA|aB$

$S \rightarrow ASD|SD|AS|aB|a$

$S \rightarrow ASA|SA|AS|aB|a|S$

$A \rightarrow B|S$

$B \rightarrow b$

eliminar unitarios:

$S \rightarrow A \rightarrow B$ $P_0 = \{(S,S), (A,A), (B,B)\}$

$(A,A): A \rightarrow B$ (A,B)

$(A,A): A \rightarrow S$ (A,S)

$(A,A): -$

$(A,S): ASA|AS|SA|aB|a$

$(A,B): b$

$S \rightarrow ASA|AS|SA|aB|a$

$A \rightarrow ASA|AS|SA|aB|a|b$

$B \rightarrow b$ 1

3. eliminar inútiles:

$T: \{a,b\}$

$Y_1: \{S\}$

$Y_4: \{S,A,B,a,b\}$

$W_1: \{S,A,B,a,b\}$

$Y_2: \{S,A,a,b\}$

$S \rightarrow ASA|AS|SA|aB|a$

$W_2: \{S,A,B,a,b\}$

$Y_3: \{S,A,B,a,b\}$

$A \rightarrow ASA|AS|SA|aB|a|b$

$B \rightarrow b$

4. Chomsky

$C_1 = AS$ $T_1 = a$ $T_2 = b$

$S \rightarrow C_1A|AS|SA|T_1B|T_1$

$A \rightarrow C_1A|AS|SA|T_1B|T_1|T_2$

$B \rightarrow T_2$ 1